Problem Set 4E

Collinearity

Do these after the slides for Section 4E. Textbook Exercise 4E. Answers are on the last page. Show full working in your exercise book.

Skills

1 Show that $A(1,-1,2)$, $B(3,2,1)$ and $C(7,8,-1)$ are collinear, and find $AB : BC$.

2 Are $A(0,1,2)$, $B(1,3,5)$ and $C(2,5,9)$ collinear? Justify your answer.

3 $A(2,3,-1)$, $B(4,0,1)$ and $C(8,m,n)$ are collinear. Find m and n.

4 $A(-1,2,5)$ and $B(7,-2,1)$. Find the point P on AB with $AP : PB = 3 : 1$.

5 A, B and C have position vectors $\mathbf{a}$, $\mathbf{b}$ and $3\mathbf{b} - 2\mathbf{a}$. Show that A, B and C are collinear and find $AB : BC$.

Exam style

6 $OABC$ is a parallelogram with $\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{OC} = \mathbf{c}$. M is the midpoint of AB and X lies on OB with $\overrightarrow{OX} = \frac{2}{3}\overrightarrow{OB}$. Show that C, X and M are collinear and find $CX : XM$.

7 In triangle OAB, P lies on OA with $OP : PA = 3 : 1$ and Q is the midpoint of AB. The lines OQ and BP meet at R. Find $\overrightarrow{OR}$ in terms of $\mathbf{a}$ and $\mathbf{b}$.

8 $A(1,2,3)$ and $B(3,-2,5)$. The point C lies on the line AB, beyond B, with $AC = 3AB$. Find the coordinates of C.

Problem Set 4E: answers

1 $\overrightarrow{BC} = 2\overrightarrow{AB}$ with B common; $AB : BC = 1 : 2$

2 No: $\overrightarrow{AB} = \mathbf{i} + 2\mathbf{j} + 3\mathbf{k}$ and $\overrightarrow{AC} = 2\mathbf{i} + 4\mathbf{j} + 7\mathbf{k}$ are not parallel

3 $\overrightarrow{AC} = 3\overrightarrow{AB}$: $m = -6,\ n = 5$

4 $P(5, -1, 2)$

5 $\overrightarrow{AC} = 3(\mathbf{b} - \mathbf{a}) = 3\overrightarrow{AB}$; $AB : BC = 1 : 2$

6 $\overrightarrow{CX} = \frac{2}{3}\mathbf{a} - \frac{1}{3}\mathbf{c}$ and $\overrightarrow{CM} = \mathbf{a} - \frac{1}{2}\mathbf{c} = \frac{3}{2}\overrightarrow{CX}$; $CX : XM = 2 : 1$

7 $\overrightarrow{OR} = \frac{3}{7}\mathbf{a} + \frac{3}{7}\mathbf{b}$

8 $C(7, -10, 9)$